

LESSON 6-2 • SESSION 2 • PRACTICE • FORM B

Dividing with Mixed Numbers

Learning target – I can divide mixed numbers by writing them as improper fractions first. • 6.NS.A.1

REMEMBER

- Write each mixed number as an **improper fraction**:
 $2\frac{3}{4} = (2 \times 4 + 3)/4 = 11/4$.
- Then **multiply by the reciprocal** of the divisor.
- Simplify. Write an improper answer as a **mixed number**.

WORKED EXAMPLE

A ribbon is $6\frac{3}{4}$ meters long. Each bow uses $2\frac{1}{4}$ meters. How many bows can be made?

Step 1 Convert: $6\frac{3}{4} = 27/4$ and $2\frac{1}{4} = 9/4$.

Step 2 Multiply by the reciprocal: $27/4 \times 4/9 = 108/36$.

Step 3 Simplify: $108/36 = 3$. Check: $3 \times 2\frac{1}{4} = 6\frac{3}{4}$.

Answer 3 bows

WORDS mixed number (*número mixto*) • improper fraction (*fracción impropia*) • simplify (*simplificar*)

PRACTICE

show your work • label units

1. Divide. Write each quotient in simplest form.

a. $4\frac{2}{3} \div 1\frac{1}{6} =$ _____ b. $2\frac{5}{8} \div 7/8 =$ _____ c. $6\frac{1}{4} \div 1\frac{1}{4} =$ _____

d. $3\frac{1}{3} \div 1/4 =$ _____ e. $2\frac{1}{4} \div 4\frac{1}{2} =$ _____ f. $3\frac{3}{5} \div 6 =$ _____

2. **Paper and Sand.** Ms. Chen has a $5\frac{1}{4}$ -foot strip of poster paper and $8\frac{3}{4}$ pounds of sand.

- a. She cuts the paper into $3/8$ -foot sections. How many sections? _____ sections
- b. Each tub holds $1\frac{3}{4}$ pounds of sand. How many tubs? _____ tubs

Show your work.

3. **Paint Coats.** A painter has $6\frac{2}{3}$ liters of paint. Each coat uses $1\frac{2}{3}$ liters. How many coats can he paint?

Hint $6\frac{2}{3} = 20/3$ and $1\frac{2}{3} = 5/3$.

Show your work.

Answer _____ coats

4. **Yarn Scarves.** Ana has $13\frac{1}{2}$ yards of yarn. Each scarf needs $2\frac{1}{4}$ yards. How many scarves can she knit?

Hint Write $13\frac{1}{2}$ as $27/2$.

Show your work.

Answer _____ scarves

5. **Sugar Scoops.** A jar holds $5\frac{1}{2}$ cups of sugar. Elena says it fills $2\frac{3}{4}$ half-cup scoops. Justify or correct her.

Hint More scoops or fewer scoops than cups?

Show your work.

Answer _____ scoops

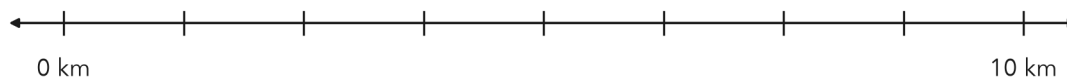
6. **Door Trim.** A shop has $14\frac{1}{4}$ ft of trim. A door uses $4\frac{1}{4}$ ft. The rest is cut into $1\frac{1}{4}$ -ft pieces.

Hint Subtract first, then divide by $5/4$.

- a. How much trim is left? _____ ft
- b. How many pieces? _____ pieces

Show your work.

- 7. Water Stations.** A race course is 10 km long. Water stations are placed every $2\frac{1}{2}$ km, starting at the $2\frac{1}{2}$ km point. Ticks are every $1\frac{1}{4}$ km.



- How many $2\frac{1}{2}$ -km spacings fit along the course? _____ spacings
- How many stations are placed, counting the one at the finish? _____ stations
- Why are the two counts the same? _____

- 8. Punch Pitchers.** A jug holds $8\frac{3}{4}$ cups of punch. Each pitcher holds $1\frac{1}{4}$ cups. How many pitchers can be filled? Check by multiplying.

Hint Check: $7 \times 1\frac{1}{4}$ should equal $8\frac{3}{4}$.

Show your work.

Answer _____ pitchers

FIND THE ERROR

explain the mistake, then fix it

- 9. Jada's Work.** Jada solved $3\frac{2}{5} \div \frac{4}{5}$. She wrote $3\frac{2}{5} = \frac{5}{5}$, then $\frac{5}{5} \times \frac{5}{4} = 1\frac{1}{4}$. Explain her mistake. Then find the correct quotient.

TEST PRACTICE

choose the best answer

- 10.** Written as an improper fraction, $4\frac{2}{3}$ is –

A $\frac{8}{3}$ **B** $\frac{14}{3}$ **C** $\frac{6}{3}$ **D** $\frac{10}{3}$

- 11.** Find $2\frac{1}{2} \div 1\frac{1}{4}$.

A $\frac{1}{2}$ **B** $2\frac{1}{8}$ **C** 2 **D** $3\frac{1}{8}$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

6 V	5 I	7 S	4 N	8 T	11 2/5 Z	1/4 X	2 3/4 Q	11 E	10 R
#2b		#3	#4	#5	#6a	#6b	#8		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____